

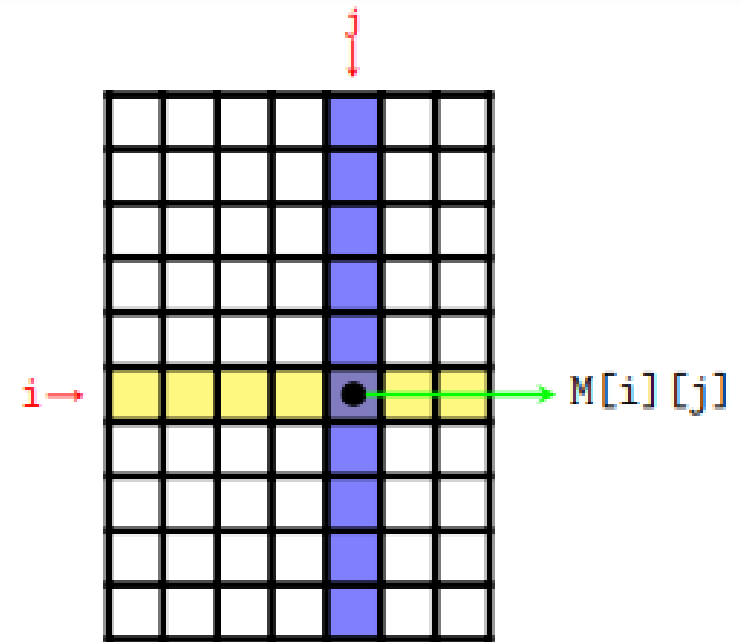
MATRIXES

❖ A matrix is declared as follows

Identifier: array $[S_{\text{maxLig}}][S_{\text{maxCol}}]$ of type;

Where,

- Identifier: Represents the name of the matrix.
 - S_{maxLig} : is the maximum number of rows.
 - S_{maxCol} : is the maximum number of columns.
 - Type: is the type of single or compound elements.
- ❖ To access the elements of a matrix, we use the operator $[]$ by specifying a position by dimension: we first specify the row number, and then the column number:



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❖ Several operations can be performed on the matrices:

Consulting Operations	Modification operations
• Displaying a matrix • Calculation of the sum, the product, the maximum, the minimum, the transpose, the inverse,... for the elements of the matrix • Check the properties of a matrix: diagonal, symmetrical, magic, etc.	• Reading (entering or filling in) a matrix • Edit elements • Items offset

❖ All these operations require the elements of the matrix to be traversed. There are several types of paths: row by row, column by column, partial, a row /column, the main diagonal/secondary diagonal, etc.

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- ❖ **Exercise 1:** Let a matrix A of size $n*m$ ($n \leq 20$, $m \leq 30$), write an algorithm that:
1. Reads (fills in) the values of matrix A with two different methods (by row and by column).
 2. Displays the maximum as well as the average of all elements of A .
 3. Constructs matrix B which represents the transpose of matrix A .
 4. Displays the sum of each column of B .
 5. Displays matrix A row by row, and matrix B column by column.

MATRIXES

❖ Exercise 1 Solution:

Algorithm Operations_Matrice1;

Var

A, B : array of [20] [30] of integer;

i, j, n, m, S, Max : integer; Moy : real;

// 1. Read Matrix A

Do

print("give n between 1 et 20 ");

scan(n);

while (n<1) or (n>20);

Do

print("give m between 1 et 30 ");

scan(n);

while (n<1) or (n>30);

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❖ Exercise 1 Solution:

// 1. Read Matrix A

//reading by row

For $i \leftarrow 0$ to $n-1$ do

 For $j \leftarrow 1$ to $m-1$ de

 Begin

 print("give an element");

 scan(A[i][j]);

 End_{For};

//reading by column

For $i \leftarrow 0$ to $m-1$ do

 For $j \leftarrow 1$ to $n-1$ de

 Begin

 print("give an element");

 scan(A[j][i]);

 End_{For};

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❖ Exercise 1 Solution:

// 2. Display the maximum as well as the average of all the elements of A

$S \leftarrow 0$; $Max \leftarrow A[0][0]$;

For $i \leftarrow 0$ to $n-1$ do

 For $j \leftarrow 0$ to $m-1$ de

 Begin

$S \leftarrow S + A[i][j]$;

 if($A[i][j] > Max$) then $Max \leftarrow A[i][j]$;

 End_{For}

$Moy \leftarrow S / (n * m)$;

Print("Moy=", Moy, " Max=", Max);

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❖ Exercise 1 Solution:

// 3. Construct matrix B which represents the transpose of matrix A.

For $i \leftarrow 0$ to $n-1$ do

 For $j \leftarrow 0$ to $m-1$ de

$B[j][i] \leftarrow A[i][j];$

// 4. Display the sum of each column of B.

For $i \leftarrow 0$ to $m-1$ do

 Begin

$S \leftarrow 0;$

 For $j \leftarrow 0$ to $n-1$ de

$S \leftarrow S + B[j][i];$

 Print("la somme des éléments de la colonne", i, " de la matrice B est :", S);

End_{For}

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❖ Exercise 1 Solution:

// 5. Display matrix A row by row, and matrix B column by column.

```
//Display by row
```

```
For i ← 0 to n-1 do
```

```
    For j ← 1 to m-1 de
```

```
        print("A[" , i, "]" [ , j, "]=", A[i][j]);
```

```
//Display by column
```

```
For i ← 0 to m-1 do
```

```
    For j ← 1 to n-1 de
```

```
        print("B[" , i, "]" [ , j, "]=", B[i][j]);
```

COURSE SOURCES

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